A Universal 8-Vertex Origami Torus Triangulation

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Abstract

An origami torus is a piecewise affine isometric embedding of a flat torus into $\boldsymbol{R}^3$. We prove that every flat torus is realized as an 8-vertex origami torus based on the triangulation with degree sequence $4, 6, 6, 6, 6, 6, 7, 7$. This provides an 8-vertex universal triangulation for origami tori. This initial version of the paper does not explore the beautiful geometry of the objects involved or the theory behind them; it just gets the job done.

1 Introduction

1.1 The Main Result

A *flat torus* is a quotient of the form $T = \boldsymbol{R}^2/\Lambda$, where Λ is a lattice of translations of $\boldsymbol{R}^2$. The lattice Λ is called the *lattice* of T. A *triangulation* $\mathcal{T}$ of a flat torus T is a decomposition of T into finitely many triangles, such that every two triangles are either disjoint, or share one vertex, or share one edge.

Two triangulations $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively of flat tori T_1 and T_2, are *combinatorially equivalent* if there is a homeomorphism $h : T_1 \to T_2$ which affinely maps each triangle of $\mathcal{T}_1$ to a triangle of $\mathcal{T}_2$. The equivalence class of $\mathcal{T}$ is called its *combinatorial type*. We will sometimes blur the distinction between a triangulation and its combinatorial type. There are 7 different combinatorial types of 8-vertex triangulations of tori. Figure 1.1 below shows the 8-vertex triangulations $\mathcal{T}(46666677)$ and $\mathcal{T}(66666666)$. We are naming these particular triangulations by their degree sequence. In each of these two cases, the degree sequence determines the combinatorial type.

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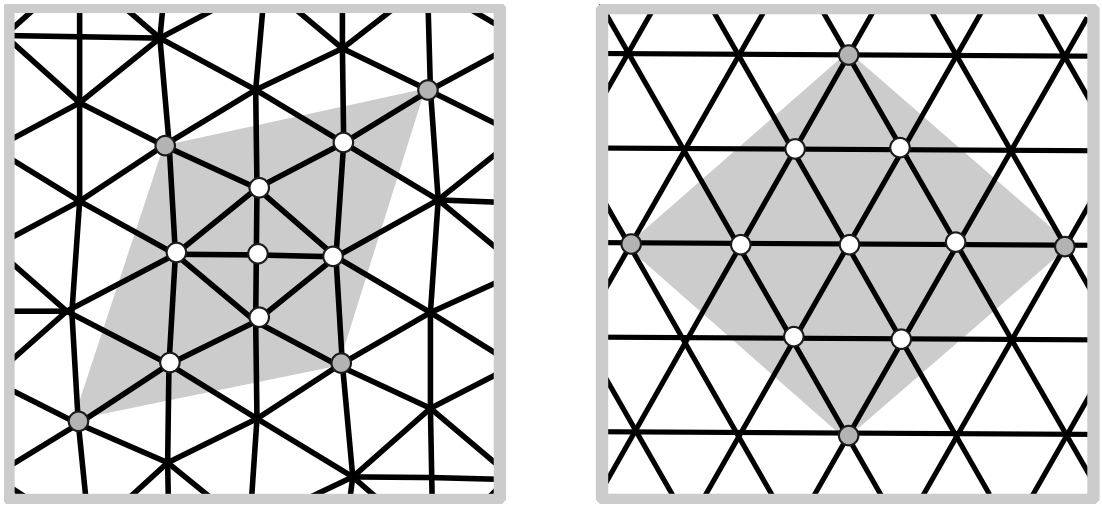


Figure 1.1: The triangulations $\mathcal{T}(46666677)$ and $\mathcal{T}(66666666)$.

An *origami torus* is an embedded topological torus in $\boldsymbol{R}^3$ that is made by fitting together finitely many triangles so that the cone angle around each vertex is 2π. Equivalently, an origami torus is the image of a piecewise affine isometric embedding $\phi : T \to \boldsymbol{R}^3$ of a flat torus T. This is to say that T has a triangulation $\mathcal{T}$ such that the restriction of ϕ to each triangle of $\mathcal{T}$ is an isometric affine map. (The specific triangulation is part of the data.) We call a lattice Λ *realized* by $\mathcal{T}$ if there exists an origami torus associated to the lattice Λ and the combinatorial type $\mathcal{T}$. We call a combinatorial type $\mathcal{T}$ *universal* if every lattice Λ is realized by $\mathcal{T}$. Here is our main result.

Theorem 1.1 (Universality) $\mathcal{T}(46666677)$ *is universal.*

The Universality Theorem says that there is a single 8-vertex combinatorial type out of which you can build an origami torus of any shape you like. Our construction yields 2 continuous families of 8-vertex origami tori based on $\mathcal{T}(46666677)$ which together realize all shapes. We do not know if a single continuous family suffices, but see Remark 2 in §2.1. Here 8 is as good as it gets: In [**S1**] it is proved that 7-vertex origami tori do not exist. See also [**S2**].

1.2 History and Context

The Universality Theorem is the culmination of a series of results about origami tori, some quite recent.

• In 1960, Y. Burago and V. Zalgaller [**BZ1**] give the first construction of origami tori.

• in 1995, Y. Burago and V. Zalgaller [**BZ2**] prove that every lattice is realized by some combinatorial type.

• In 2020, T. Tsuboi [**T**] gives explicit constructions which realize every lattice as an origami torus called a *diplotorus* [1]. P. Arnoux, S. Lelièvre, and A. Málaga [**ALM**] give an independent proof of the same result. The construction requires infinitely many combinatorial types. Diplotori were discovered by U. Brehm [**Br**] in 1978. See also [**Se**, §6].

• In 2023, F. Lazarus and F. Tallerie [**LT**] give a universal triangulation having 1217 vertices. Their paper also has an excellent discussion of the various attempts made to realize all lattices as origami tori.

• In 2025, R.E.S. [**S1**] constructs an 8-vertex origami torus and proves that a 7-vertex origami torus does not exist. See also [**S2**]. The 8-vertex origami tori in [**S1**] and [**S2**], called *pup tents*, have combinatorial type $\mathcal{T}(66666666)$ and are symmetric with respect to order 2 rotation about the Z-axis.

• In 2025, P.G.D. and R.E.S. [**DS**] prove that a lattice is realized by a pup tent provided that it is neither rectangular nor sharp rhombic. A lattice is *rectangular* if it is generated by a pair of perpendicular vectors and *sharp rhombic* if it is generated by a pair of equal-length vectors which make an angle of at most $\pi/3$ with each other. When the angle is $\pi/3$ you get the famous *hexagonal lattice*. Thus, almost every lattice is realized by a pup tent. In [**DS**] the authors speculate that perhaps the remaining lattices can be realized by 8-vertex origami tori without 2-fold symmetry.

• In 2026, F. Lander [**L**] proves that the square lattice is realized by $\mathcal{T}(66666666)$ and the hex lattice is realized by $\mathcal{T}(46666677)$. These origami tori have no symmetries.

The result in [**L**] grew out of some experiments that F.L. and S.T. did in the summer of 2026. These authors made a poster for R.E.S.'s birthday conference and, in an exciting development, their software could realize some rectangular and sharp rhombic lattices. Upon further experimentation, F.L. and S.T. discovered that the folding pattern used to realize the hexagonal lattice from $\mathcal{T}(46666677)$ turned out to be extremely robust. This fact led to the Universality Theorem.

[1] For the rectangular lattices one needs a slightly modified construction.

1.3 Proof Strategy and Philosophy

If we fix a triangulation $\mathcal{T}$, the set $O(\mathcal{T})$ of all origami tori which have $\mathcal{T}$ as the underlying combinatorial type is a semi-algebraic set. This is to say that $O(\mathcal{T})$ is cut out by algebraic equalities and inequalities. The equalities govern the intrinsic flatness conditions – i.e. the 2π angle sum – around each vertex, and the inequalities govern the embedding property.

Here is how this works. The variables are

1. A point (x, y) such that Λ is generated by $(1, 0)$ and (x, y).
2. A collection of 8 points $w_0, ..., w_7 \in \boldsymbol{R}^2$.
3. Points $p_0, ..., p_7 \in \boldsymbol{R}^3$ realizing the vertices of a member of $O(\mathcal{T})$.

The equalities mentioned above say that each relevant distance $\|p_i - p_j\|^2$ equals an expression of the form $\|w_i' - w_j'\|^2$ where w_i' and w_j' are suitable Λ-translates of w_i and w_j, chosen so that the segment joining w_i' and w_j' is isometric to the corresponding edge of $\mathcal{T}$. Various polynomial inequalities, encoding the positions of the triangles in space, determine the embedding properties.

Ideally, we would like to get explicit coordinates for the vertices of points in $O(\mathcal{T})$. This seems quite difficult, partly because there are too many choices to make. The central idea that emerged in [**DS**] is that one should try to parametrize points in the boundary $\partial O(\mathcal{T})$. This boundary is a stratified space, and the points in it correspond to *gracefully immersed origami tori*. Such an object is a piecewise affine isometric map $\phi : T \to \boldsymbol{R}^3$ which is not an embedding but which is approximated arbitrarily closely by piecewise affine isometric embeddings.

Essentially, when we move to the boundary we allow certain faces to collapse together. Roughly speaking, the more we have collapsed the faces, the lower the dimension of the stratum we land in. For gracefully immersed origami tori, the map ϕ is very important because it records how the collapsing is done.

The collapsing operation makes certain choices for us, and gives us a better chance of finding nice parametrizations of boundary strata. In [**DS**], the authors found a 2-dimensional stratum which was essentially a copy of a fundamental domain for the modular surface, the hyperbolic orbifold that parametrizes lattices modulo similarity. Aside from one square root, the parametrization is rational. Once the authors found this parametrization, they found an algebraic path from the boundary stratum back into $O(\mathcal{T})$. Certain degeneracies prevented the path from realizing rectangular and sharp rhombic lattices. The strategy here is related but we get everything.

Step 1: In the present setting, where we work with $\mathcal{T} = \mathcal{T}(46666677)$, we give an explicit rational parametrization of a 5-dimensional boundary stratum. This stratum is very special: The corresponding gracefully immersed origami tori lie entirely in the plane! This is the first step of our construction. We call these origami tori *flattened origami tori*. We will give formulas in §2.

Step 2: We think of our origami tori as living in $\boldsymbol{R}^2 \times \{0\}$. Once we have parametrized our flattened origami tori, we will establish (to a sufficient extent) that they have a certain *lifting property*. This means that we will keep the XY-coordinates the same and add suitable Z-coordinates $(tz_0, ..., tz_7)$, depending on the example, so that the flattened origami torus becomes an embedded polyhedral torus. What we mean by *to a sufficient extent* is that we will not establish this lifting property for the full 5-dimensional family, but only for a 2-dimensional subset which is sufficient to realize every similarity class of lattice. We carry this out in §3.

Step 3: Our lifting path does not quite give us origami tori on the nose. To correct the lifting path, we will use a strategy like that used in [**L**]. We prove a transversality result, essentially the Implicit Function Theorem, which shows that we can recover perfect flatness and get the embedding without changing the modulus. (The approach in [**DS**] is related but treats the modulus differently.) We do this in §4.

1.4 Organization

In §2-4 we will give the main proof along the lines of the steps above. On several occasions, we will use, as a black box, the positivity of certain polynomials on certain domains. in §5 we will explain how we certify this positivity.

1.5 Computer Calculations

We perform our calculations in Mathematica [**W**]. These calculations are all fairly short and easy to reproduce. One file contains everything. Here are two places to get the file.

http://www.math.brown.edu~res/Java/UniversalityProof

https://github.com/FabianLander/universal-8-vertex-origami-torus-triangulation

1.6 Artificial Intelligence Use

We used Artificial Intelligence in a moderate way. The conceptual framework for the proof is ours, and evolved from the papers [**S1**], [**DS**], and [**L**]. At various stages, ChatGPT 5.6 implemented some of our search algorithms and optimized some of our code. Perhaps needless to say, the AI also helped format equations, provide references, and proofread the paper. Also, we mention that ChatGPT had an excellent idea for simplifying the angle-derivative formulas for the Jacobian calculation in §4.

1.7 Acknowledgements

R.E.S. would like to thank the I.H.E.S for their support, in the form of the Gretchen and Barry Mazur Chair, during the writing of this paper.

2 The Flattened Tori

2.1 The 5 Parameter Family

We are going to describe a 5-parameter family of flattened origami tori. We think of this family as a fiber bundle over a base space coordinatized by (x, y). The fiber is parametrized by variables (u, k, λ) which satisfy certain algebraic constraints.

The Lattice: The variables (x, y) correspond to a lattice

$$\Lambda = \boldsymbol{Z}(1, 0) + \boldsymbol{Z}(x, y). \tag{1}$$

That is, Λ is the $\boldsymbol{Z}$-lattice generated by $(1, 0)$ and (x, y). Here we are suppressing the dependence of Λ on x and y.

The Parameters: The base space will be $\mathcal{M}$, the modular domain

$$x \in [0, 1/2], \qquad y \geq \sqrt{1 - x^2}. \tag{2}$$

When we think of $\mathcal{M}$ as a subset of the hyperbolic plane $\boldsymbol{H}^2$ in the upper half-plane model, $\mathcal{M}$ is the $(2, 3, \infty)$ hyperbolic triangle. Up to scaling and reflection, every lattice is generated by vectors $(1, 0)$ and $(x, y) \in \mathcal{M}$. The family is defined for a wider range of values of (x, y), but unless otherwise stated we only care about the case $(x, y) \in \mathcal{M}$.

The Flattened Origami Torus: Here are the points of the corresponding flattened torus $T(x,y,u,k,\lambda)$. In listing these vertices, we find it useful to also define the auxiliary variables:

$$a = u - x - k + 1/2, \qquad b = 1 - x - 2k. \tag{3}$$

We also find it convenient to scale the points by $2y$ when we list them.

$$\begin{aligned}
2yp_0 &= (0,0,0)\\
2yp_1 &= (2uy,\ 2u(b+2k/\lambda),\ 0)\\
2yp_2 &= ((2u-1)y,\ (1-2u)x,\ 0)\\
2yp_3 &= (-2y(k+b\lambda),\ \lambda(y^2-b^2)-2bk,\ 0)\\
2yp_4 &= ((2x+2k-1)y,\ y^2+bx,\ 0)\\
2yp_5 &= ((2u-1)y,\ y^2+b(b+2u-1),\ 0)\\
2yp_6 &= (2y(u-k),\ 2b(u-k),\ 0)\\
2yp_7 &= (2uy,\ y^2+b^2+2bu-2a/(1-\lambda),\ 0)
\end{aligned}$$

The Flat Torus: We will describe 16 triangles. We get a tiling of $\boldsymbol{R}^2$ by taking the orbit of these triangles under the action of the lattice Λ defined above. The quotient then gives us our 8 vertex triangulation of $\boldsymbol{R}^2/\Lambda$. Here are the triangles

$$\begin{array}{llll}
012 & (w_0,w_1,w_2) & 167 & (w_1,w_6,w_7)\\
031 & (w_0,w_3,w_1) & 264 & (w_2+e,w_6,w_4+e-\tau)\\
024 & (w_0,w_2,w_4-\tau) & 257 & (w_2+e,w_5+e,w_7)\\
043 & (w_0,w_4-\tau,w_3) & 276 & (w_2+e,w_7,w_6)\\
152 & (w_1,w_5,w_2) & 347 & (w_3+\tau,w_4,w_7)\\
136 & (w_1,w_3,w_6) & 356 & (w_3,w_5+e-\tau,w_6)\\
145 & (w_1,w_4,w_5) & 375 & (w_3+\tau,w_7,w_5+e)\\
174 & (w_1,w_7,w_4) & 465 & (w_4+e-\tau,w_6,w_5+e-\tau)
\end{array} \tag{4}$$

Here $e=(1,0)$ and $\tau=(x,y)$. Setting $\pi(r,s,0)=(r,s)$, we have

$$\begin{aligned}
w_j &= \pi(p_j) \qquad j \neq 3,6\\
2yw_3 &= (2ky,\ -2bk-\lambda(b^2+y^2))\\
2yw_6 &= (2y(u+k),\ 2b(u-k))
\end{aligned} \tag{5}$$

Preliminary Isometric Mapping: Let W_j denote the jth triangle above. We call the set $\{W_0,...,W_{15}\}$ the *fundamental triangles*. Let P_j denote the corresponding triangle whose vertices are the p points. Let $\phi : \bigcup W_j \to \bigcup P_j$ be the piecewise affine map from the one set of 16 triangles to the other.

Lemma 2.1 *ϕ is an isometry on each of the 16 triangles.*

Proof: We compute this exactly with the file **Universality Proof**. ♠

Remarks: (1) Within the 5-parameter family there are some algebraic constraints on the fiber (u, k, λ) which suffice to ensure that the triangles above have positive area:

$$u, k, a, \lambda/2 \in (0, 1/2), \qquad \frac{a}{1-\lambda} + \frac{2ku}{\lambda} < b + 2k(x+k). \tag{6}$$

We will never need to check or work with with these constraints.
(2) The constraint $a \in (0, 1/2)$, or equivalently $x + k - u \in (0, 1/2)$, forces $k - u > 0$ when $x = 0$ and $k - u < 0$ when $x = 1/2$. So, any continuous family of examples where x ranges from 0 to $1/2$ must include a case where $k = u$. When $k = u$ we have $p_0 = p_6$, and it seems impossible to perturb these examples to embedded polyhedral tori by adding third coordinates. This may be an obstacle to improving our result so that one continuous family of origami tori realizes all lattice shapes.

2.2 The Special Families

We now specialize to two 2-dimensional sub-families:

$$\text{Family 1}: \quad x \in \left[0, \frac{1}{6}\right], \qquad u = \frac{3}{128}, \qquad k = \frac{11}{32}, \qquad \lambda = \frac{13}{32}.$$

$$\text{Family 2}: \quad x \in \left[\frac{1}{6}, \frac{1}{2}\right], \qquad u = \frac{x}{2} - \frac{1}{2048}, \qquad k = \frac{1}{16}, \qquad \lambda = \frac{1}{2}.$$

Both families are defined in the entire upper half-plane, but we consider Family d on *Domain* d for $d = 1, 2$. These domains are respectively $\mathcal{M}|_{[0,\frac{1}{6}]}$ and $\mathcal{M}|_{[\frac{1}{6},\frac{1}{2}]}$. Here $\mathcal{M}|_{[a,b]}$ is the subset of $\mathcal{M}$ satisfying $x \in [a, b]$. The cutoff of $1/6$ is a bit arbitrary; Anything in $[.14, .1667]$ works.

Lemma 2.2 *For all parameters in Domain d, every fundamental triangle in Family d has positive area. Here $d = 1, 2$.*

Proof: For each triangle in Family d we compute that $\det(A - C, B - C)$ does not vanish in Domain d using the strip dominance method in §5. Here A, B, C are the vertices of the triangle. ♠

Figure 2.1 shows the triangulated flat torus from Family 1 associated to the parameters $(x, y) = (0, 1)$, the square lattice. The white triangles are translates of the fundamental triangles by short vectors in Λ.

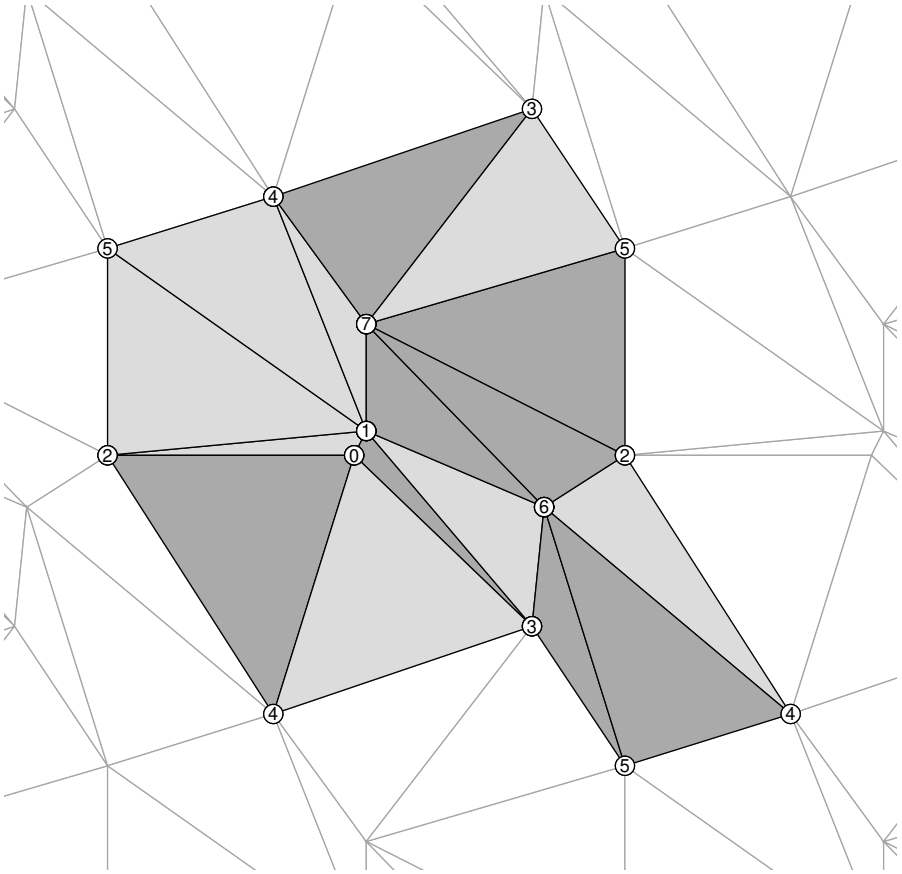


Figure 2.1: The triangulated flat torus associated to the square lattice.

From the vertex labeling you can see how the action of Λ identifies the edges along the boundary to make the torus.

1. $e = (1, 0)$ maps edges 42 and 25 on the left to edges 42 and 25 on the right.
2. $\tau = (0, 1)$ maps edges 43 and 35 on the bottom to edges 43 and 35 on the top.
3. $e - \tau$ maps the top left edge 45 to the bottom right edge 45.

This is not just a visual check; it comes from the formulas in Equation 4. For instance, you can see the action of $e - \tau$ in Item 3 by comparing the formulas for triangle 145 and triangle 465.

The light shading indicates triangles for which the map ϕ above preserves orientation. The binary sequence 0110000010111101 carries all the information about the orientations. The jth bit is a 0 iff the corresponding triangle sign is $+$. Referring to Equation 4, the ordering is such that triangles $0 - 7$ are on the left.

Figure 2.2 shows how the corresponding origami torus is built triangle by triangle, with an ordering compatible with the lift we give in the next chapter.

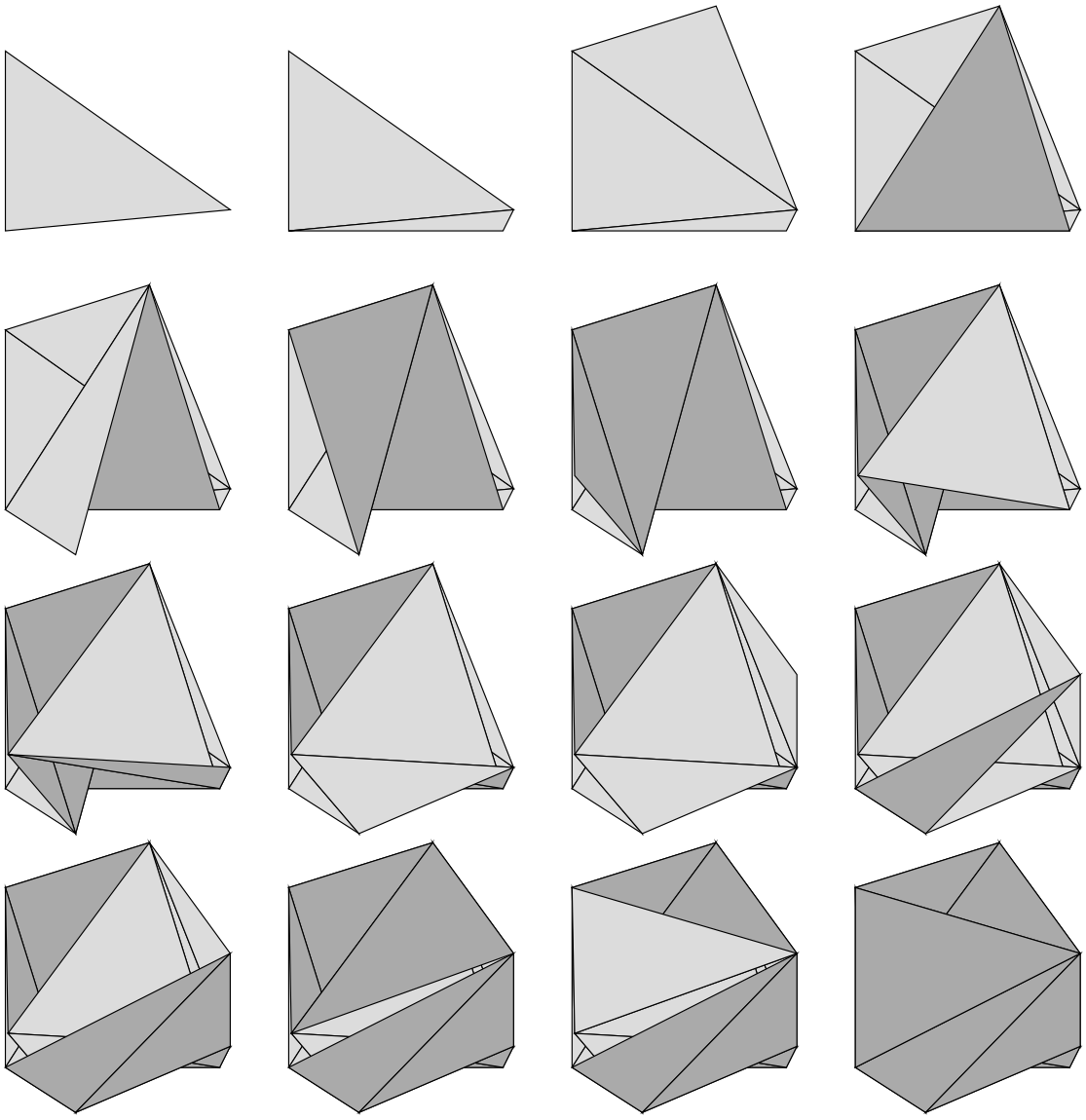


Figure 2.2: The flattened origami torus associated to the square lattice.

Figures 2.3 and 2.4 show the same thing as for the torus in Family 2 with the parameter $(8/17, 1)$. A more natural choice would be the parameter $(1/2, \sqrt{3}/2)$ for the hex lattice, but in this case the picture does not look great. Two of the points are very close together for the hex lattice picture.

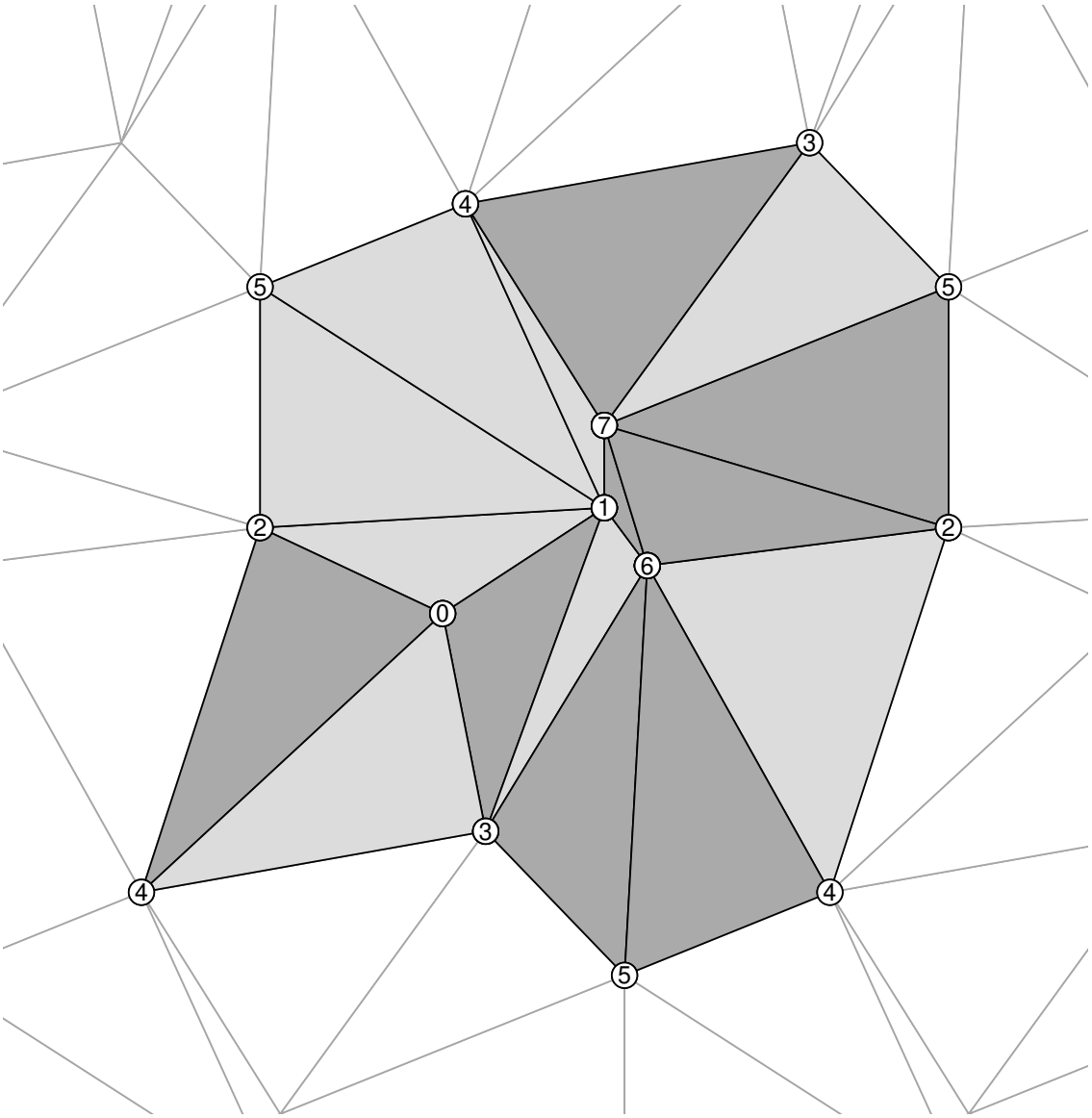


Figure 2.3: The triangulated flat torus associated to $(8/17, 1)$.

In Figure 2.3, the side-pairings work exactly as in Figure 2.1.

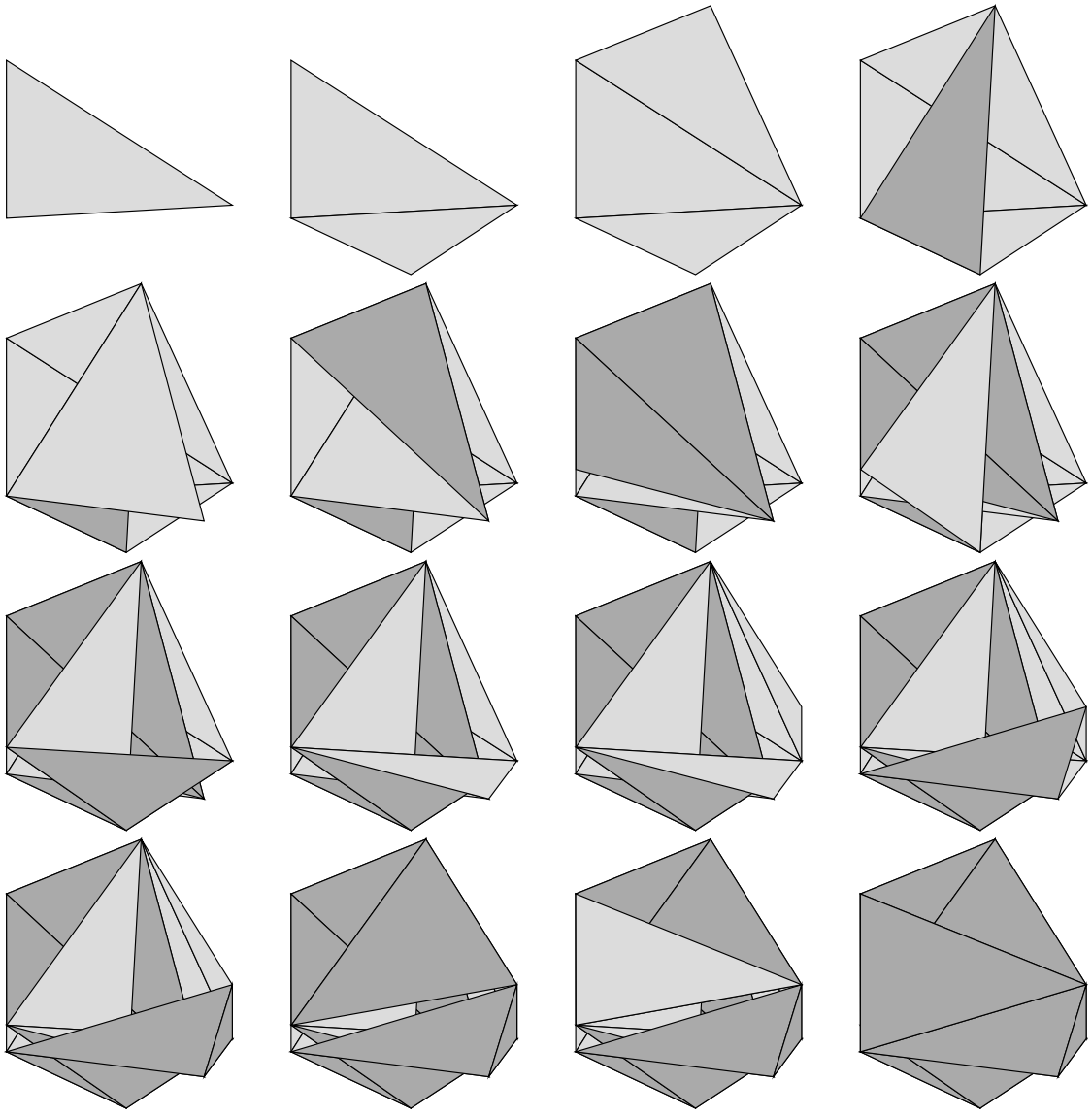


Figure 2.4: The flattened origami torus associated to $(8/17, 1)$.

2.3 Producing the Flattened Origami Tori

Let $F = \bigcup W_j$ and $\Psi = \bigcup P_j$. We have a map $\phi : F \to \Psi$ which is an isometry on each triangle. For each of the figures we plotted above, F is an embedded solid polygon and the three elements $e, \tau, e - \tau$ of Λ pair the sides of F. Let T be the

quotient of F by this side-pairing. T is a topological torus. Because the side-pairings are translations, we have the familiar result that T is a flat torus. This picture persists for an open set of parameters in Domain d, for $d = 1, 2$.

As the parameters vary in Domain d, Lemma 2.2 tells us that the triangles retain their orientations and never degenerate. Also, all the angles vary analytically. Hence, if we form the abstract topological space by gluing together the triangles in the same pattern as we do when we know that F is embedded, we get a topological torus whose cone angles vary analytically. Since these cone angles equal 2π on an open set, they are always 2π. Hence, our torus T is always a flat torus. (The expert reader may enjoy proving the unnecessary fact that F is always an embedded polygon and a fundamental domain for Λ.) Furthermore, ϕ respects all the gluings needed to make T and therefore gives us a flattened origami torus $\phi : T \to \Psi$ in all cases.

3 The Lifting Property

We think of $\boldsymbol{R}^2$ as the subset $\boldsymbol{R}^2 \times \{0\} \subset \boldsymbol{R}^3$. Let $\pi : \boldsymbol{R}^3 \to \boldsymbol{R}^2$ be the projection map. Let $\phi : T \to \boldsymbol{R}^2$ be a flattened origami torus. We say that a *lift* of ϕ is a piecewise affine embedding $\widehat{\phi} : T \to \boldsymbol{R}^3$ such that $\pi \circ \widehat{\phi} = \phi$. The data is a list $(z_0, ..., z_7)$ of third coordinates for the 8 vertices of $\phi(T)$. Here are our lifts, respectively for Family 1 and for Family 2.

$$0 \quad 0 \quad -2 \quad 3 \quad 1 \quad -2 \quad 2 \quad 1024 + 1024y^2 \tag{7}$$

$$0 \quad 0 \quad -4 \quad -2 \quad 2048 \quad -4 \quad 1816x - 229 \quad 16384 + 16384y^2 \tag{8}$$

In this chapter we prove that our lifts are all embedded. We call a polyhedral torus *flexed* if no four of its vertices are coplanar.

Lemma 3.1 *On Domain d, every lifted polyhedral torus from Family d is flexed. Here $d = 1, 2$.*

Proof: For each quadruple $q = \alpha, \beta, \gamma, \delta$ we use the strip dominance method from §5 to show $\det(p_\alpha - p_\delta, p_\beta - p_\delta, p_\gamma - p_\delta)$ does not vanish on Domain d. ♠

Let T be one of our triangulated flat tori. Let $f : T \to \boldsymbol{R}^3$ be a piecewise affine map. We assume that f is injective on each triangle of the triangulation separately and that f maps the vertices to points in general position.

Lemma 3.2 *Suppose that the vertices of a polyhedral torus Ψ are in general position. Suppose also that every edge of Ψ is disjoint from every triangle of Ψ with which it shares no vertices. Then Ψ is embedded.*

Proof: We show that any two triangles $f(\Delta_1)$ and $f(\Delta_2)$ intersect precisely in $f(\Delta_1 \cap \Delta_2)$. If Δ_1 and Δ_2 share an edge, their planes are distinct by general position. Hence they intersect exactly along their common edge. Now suppose Δ_1 and Δ_2 do not share an edge. If they have an unwanted intersection, some edge of one triangle must intersect the other triangle outside their common vertices. An edge having a vertex in common with the other triangle cannot do this: by general position, it intersects the plane of that triangle only at the common vertex. Thus the unwanted intersection must involve an edge and a triangle with no vertices in common. This contradicts our hypothesis. ♠

Now we turn to the question of how we tell when a line segment $(p_0, p_1) \subset \boldsymbol{R}^3$ is disjoint from a triangle $(p_2, p_3, p_4) \subset \boldsymbol{R}^3$. Here we are naming these objects by their vertices. If we delete p_k from the list we get a tetrahedron T_k, and we let V_k denote the volume of T_k. To be precise, we put the points of T_k in increasing order and then take the signed volume. The sign counts the orientation. In the next lemma we assume that the points are in general position because this is all we need.

Lemma 3.3 (01) *and* (234) *are disjoint provided* $\max(V_0V_1, V_2V_3, V_3V_4) > 0$.

Proof: Let $[abcd]$ denote the signed volume of the tetrahedron with vertices listed in the order a, b, c, d. If $V_1 = [0234]$ and $V_0 = [1234]$ have the same sign then (01) lies strictly on one side of the plane containing (234). This gives disjointness. If (01) intersects (234) then [0123] and [0134] and [0142] have the same sign. This fails if $V_2V_3 > 0$ or $V_3V_4 > 0$. ♠

It turns out that there are 138 pairs (e, Δ) where e is an edge of $\mathcal{T}$ and Δ is a triangle of T and $e \cap \Delta = \emptyset$. We get a proof that T is embedded if all 138 pairs satisfy the criterion above. Using the file **UniversalityProof** we check this for $(0, 1)$ in Family 1 and for $(8/17, 1)$ in Family 2. Since none of the relevant signs changes throughout the families, the same inequalities hold everywhere. Hence, in both families, all lifted polyhedral tori are embedded.

Remark: We found these lifts by trial and error. They are really delicate. For instance, if you replace 229 by either 228 or 230 the lift is not generally embedded.

4 The Mapping Result

4.1 The Main Calculation

In this chapter we finish off the proof of the Universality Theorem using the Inverse Function Theorem. This approach is closely related to the approach in [**L**].

Let $\phi : T \to \Psi \subset \boldsymbol{R}^2 \times \{0\}$ be one of our flattened origami tori. We set $\Psi = \phi(T)$. Given a 7-vector $\epsilon = (\epsilon_0, ..., \epsilon_6) \in \boldsymbol{R}^7$ let $\Psi(\epsilon)$ be the result of replacing the vertex (x_i, y_i) of Ψ by $(x_i + \epsilon_i, y_i)$. We do not touch vertex 7, the last vertex. We let $\delta = (\delta_0, ..., \delta_6)$ denote the 7-vector of cone angle deficits of the perturbed triangulated torus $\Psi(\epsilon)$. Here $\delta = 2\pi - \theta$, where θ is the cone angle. We do not explicitly consider δ_7 but we note that $\sum \delta_k = 0$ by the Gauss-Bonnet Theorem.

For Family d, define

$$J_d = \det \left. \frac{\partial \delta}{\partial \epsilon} \right|_{\epsilon=0}. \tag{9}$$

This is the determinant of the matrix of partial derivatives of the function $\epsilon \to \delta$.

Lemma 4.1 *The function J_d does not vanish on Domain d.*

Proof: We use the tweaked method of strip dominance from §5 to certify this. ♠

Remark: Our Jacobian calculation is straightforward except for one fine point which we discuss. The formula for the angle α between $V = (v_1, v_2)$ and $W = (w_1, w_2)$ is

$$\alpha = \arccos\left(\frac{V.W}{\|V\|\|W\|}\right).$$

This formula is far from rational, but for the derivatives we have e.g.

$$\frac{\partial \alpha}{\partial v_1} = \sigma \frac{v_2}{v_1^2 + v_2^2}, \qquad \sigma = \text{sign}(\det(V, W)). \tag{10}$$

Here is the crucial observation. Since the triangles in our families are non-degenerate everywhere, and the domains are connected, we can determine all the σ values once and for all by looking at a single example. Both families have the same sequence of signs, so we only need to make one check. This gives us fairly simple rational expressions for J_d with respect to Family d. Equation 10 also justifies why these derivatives are entirely defined on the domains we test them on. Their denominators never vanish.

4.2 The Pseudo Modulus Function

Our lifted tori might not be origami tori. They can have non-trivial cone points. In this section we extend the notion of modulus to a more general *pseudo modulus* for polyhedral tori which are not necessarily flat or embedded. We always assume that we have the same underlying triangulation $\mathcal{T}(46666677)$ on our flat torus T.

We first distinguish some subsets of T which play an important role in our construction. First of all, we choose a basepoint w_1. Next, we single out the loops $\lambda_1 = 1621$ and $\lambda_2 = 1731$. These loops generate the fundamental group of $\pi_1(T, w_1)$. Next we single out a triangulated annulus $A_j \subset T$ that has λ_j in its boundary. In Figure 4.1, the annulus A_1 is dark-to-medium shaded. The annulus A_2 is medium-to-fairly light shaded. The set $A_1 \cap A_2$ consists of 3 medium-shaded triangles.

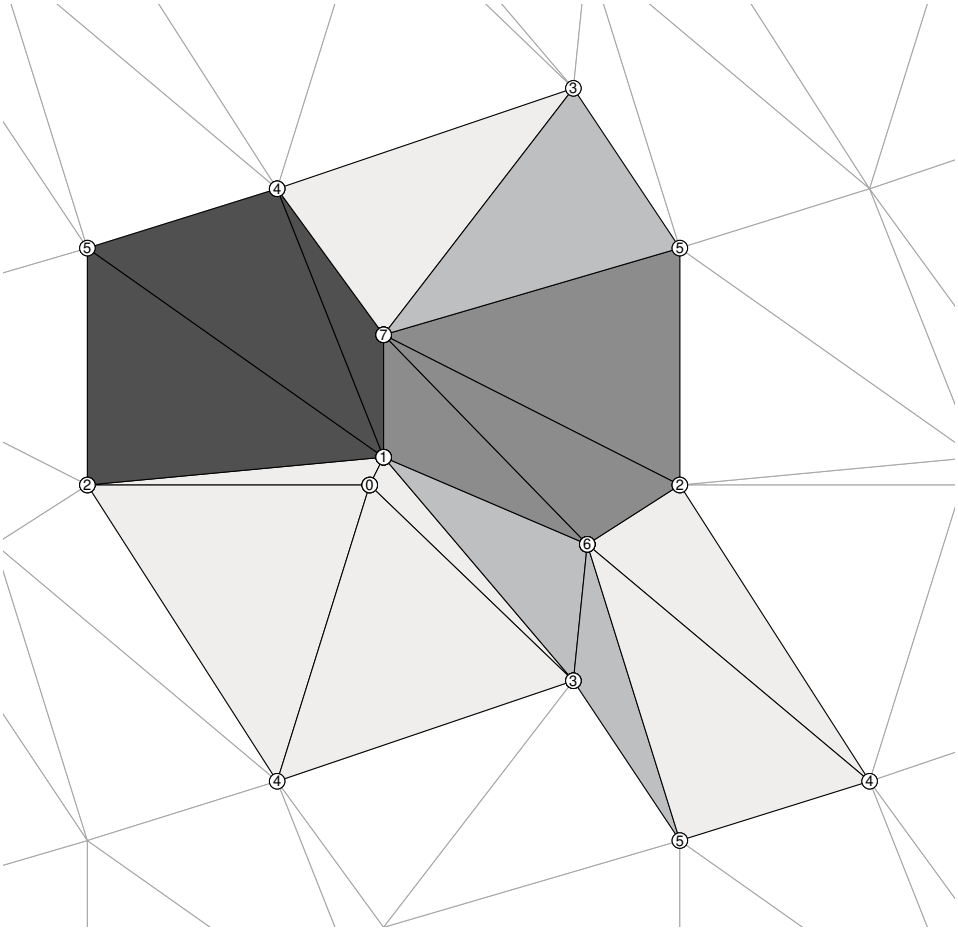


Figure 4.1: The annuli A_1 and A_2

Let $\phi : T \to \Psi \subset \boldsymbol{R}^3$ be a polyhedral torus. ϕ is piecewise affine and injective on each triangle, but not necessarily globally injective. ϕ lifts to a map $\widetilde{\phi} : \boldsymbol{R}^2 \to \Psi$. We change the metric on $\boldsymbol{R}^2$ so that $\widetilde{\phi}$ is an isometry on each triangle. We call the resulting space X. The space X is an infinite flat cone surface. When Ψ is embedded, X is the universal metric cover of Ψ. When Ψ is flat, X is isometric to $\boldsymbol{R}^2$. In general, X is homeomorphic to $\boldsymbol{R}^2$ but may have non-trivial cone points. Now we define some subsets of X. Figure 4.2 shows these subsets schematically.

- Let v be a lift of vertex w_1.
- Let Δ be the lift, of the triangle $(1, 6, 7)$, that contains v.
- For $j = 1, 2$ let L_j be the lift of λ_j that contains v. This is a bi-infinite path.
- For $j = 1, 2$, let N_j be the lift, of annulus A_j, that contains Δ. This is a bi-infinite strip.
- Let $v_1 \in L_1$ be the lift of w_1, separated from v by 3 edges, so that the directed path $\overrightarrow{vv_1}$ has N_1 on its left. Define $v_2 \in L_2$ similarly, with *right* replacing *left*.

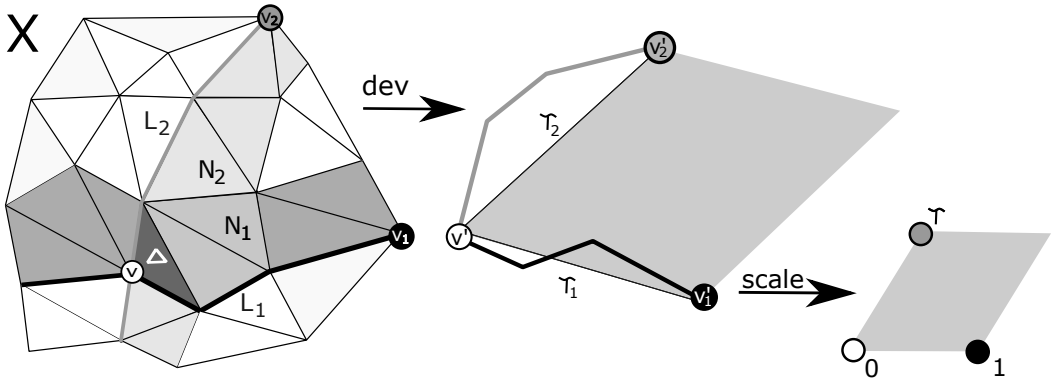


Figure 4.2: Definition of the pseudo modulus. We have set $v' = \mathrm{dev}(v)$, etc.

Since $N_1 \cup N_2$ is contractible and its interior contains no vertices, the (oriented) developing map $\mathrm{dev} : N_1 \cup N_2 \to \boldsymbol{R}^2$ is a well-defined local isometry. Define

$$\tau = \frac{\tau_2}{\tau_1}, \qquad \tau_j = \mathrm{dev}(v_j) - \mathrm{dev}(v). \tag{11}$$

We identify $\boldsymbol{R}^2$ with $\boldsymbol{C}$ in the usual way so that division makes sense. We call τ the *pseudo modulus* of Ψ. When our map $\phi : T \to \Psi$ is a piecewise affine isometry, the approach above recovers the modulus of T. In general, everything in our construction varies smoothly with the polyhedral torus because all choices are made uniformly in advance.

4.3 The End of the Proof

We want an origami torus with modulus $\zeta \in \mathcal{M}$. Say ζ belongs to Domain d for some $d \in \{1,2\}$. Family d and its lifts are defined in the entire upper half plane. Moreover, there is an open disk U, centered on ζ, for which Lemmas 2.2 and 3.1, which certify open conditions, and the analytic continuation argument in §2.3, remain true. Now we describe a 3 step process, with Steps 2 and 3 commuting.

1. We choose $\xi \in U$ and let $\Psi(\xi)$ be the corresponding flattened origami torus.
2. We perturb the X-coordinates of the vertices of $\Psi(\xi)$ to arrive at a new (not necessarily flat) polyhedral torus $\Psi(\xi,\epsilon) \subset \boldsymbol{R}^2 \times \{0\}$.
3. We scale the lift data for $\Psi(\xi)$ by t and apply it to $\Psi(\xi,\epsilon)$. This gives $\widehat{\Psi}(\xi,\epsilon,t)$.

Because Lemma 3.1 certifies open conditions, $\widehat{\Psi}(\xi,\epsilon,1)$ is embedded for sufficiently small ϵ. By affine scaling, $\widehat{\Psi}(\xi,\epsilon,t)$ is embedded for all sufficiently small ϵ and all $t \neq 0$. We call this fact the *Persistence Principle*.

Let $\mathbf{0} = (0,0,0,0,0,0,0)$. With (ξ,ϵ,t) as in the 3-step process above, let

$$F(\xi,\epsilon,t) = (\tau,\delta,t), \tag{12}$$

Here δ is the angle-deficit vector of $\widehat{\Psi}(\xi,\epsilon,t)$, and τ is its pseudo modulus. F is smooth and defined on an open set $U \times V \times W$ where $U \subset \boldsymbol{R}^2$ is as above, and $V \subset \boldsymbol{R}^7$ and $W \subset \boldsymbol{R}$ are open sets containing $\mathbf{0}$ and 0 respectively. Also, $F(\xi,\mathbf{0},0) = (\xi,\mathbf{0},0)$ for $\xi \in U$.

Lemma 4.2 *The differential dF evaluated at $(\zeta,\mathbf{0},0)$ is invertible.*

Proof: We evaluate all derivatives at $(\zeta,\mathbf{0},0)$. Note that t is independent of all other variables. Also, $\partial\tau/\partial\xi = I_2$ and $\partial\delta/\partial\xi = 0$ because $F(\xi,\mathbf{0},0) = (\xi,\mathbf{0},0)$. Hence

$$dF = \begin{pmatrix} \frac{\partial\tau}{\partial\xi} & \frac{\partial\tau}{\partial\epsilon} & \frac{\partial\tau}{\partial t} \\ \frac{\partial\delta}{\partial\xi} & \frac{\partial\delta}{\partial\epsilon} & \frac{\partial\delta}{\partial t} \\ \frac{\partial t}{\partial\xi} & \frac{\partial t}{\partial\epsilon} & \frac{\partial t}{\partial t} \end{pmatrix} = \begin{pmatrix} I_2 & B & C \\ 0 & A & D \\ 0 & 0 & 1 \end{pmatrix}, \qquad A = \frac{\partial\delta}{\partial\epsilon}, \qquad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{13}$$

Regardless of B, C, D, we have $\det(dF) = \det(A) = J_d \neq 0$ by Lemma 4.1. ♠

By Lemma 4.2 and the Inverse Function Theorem, F is a local diffeomorphism in a neighborhood of $(\zeta, \mathbf{0}, 0)$. Using the local inverse of F, we let

$$\Upsilon_t = \widehat{\Psi}(\zeta', \epsilon', t), \qquad (\zeta', \epsilon', t) = F^{-1}(\zeta, \mathbf{0}, t). \tag{14}$$

This is defined for t sufficiently small, and we take $t \neq 0$. Now, Υ_t is flat by definition, has pseudo modulus ζ by definition, and is embedded by the Persistence Principle. Hence Υ_t is an origami torus. Since Υ_t is an origami torus, its modulus equals its pseudo modulus, namely ζ. This completes the proof of the Universality Theorem.

5 Positivity Certificates

We say that a rational function $f = P/Q$ is *nice* on a domain if neither of the polynomials P nor Q vanishes on the domain. In this section we explain how we certify that all the rational functions in Lemmas 2.2, 3.1, and 4.1 are nice on the domains associated to them. This is overkill, but there is no harm in it. For Family 1 the domain is $\mathcal{M}|_{[0,\frac{1}{6}]}$ and for Family 2 the domain is $\mathcal{M}|_{[\frac{1}{6},\frac{1}{2}]}$.

Rational Parametrization: Let A be the arc of the circle $y = \sqrt{1-x^2}$ lying above the interval $[x_0, x_1] \subset [0,1]$. Define

$$x(t) = \frac{2t}{1+t^2}, \qquad y(t) = \frac{1-t^2}{1+t^2}. \tag{15}$$

We choose $r_0, r_1 \in \boldsymbol{Q} \cap [0,1]$ such that

$$x(r_0) \leq x_0 < x_1 \leq x(r_1). \tag{16}$$

Let $\Omega = [0,1] \times [0,\infty)$. The map $F : \Omega \to \boldsymbol{R}^2$ given by

$$F(t,h) = \psi(t) + (0,h), \qquad \psi(t) = (x(u), y(u)), \qquad u = (1-t)r_0 + (t)r_1 \tag{17}$$

parametrizes a superset of $\mathcal{M}|_{[x_0,x_1]}$. Here are the r-values we use:

- Family 1: For $[0, 1/6]$ we take $r_0 = 0$ and $r_1 = 43/512$.
- Family 2: For $[1/6, 1/2]$ we take $r_0 = 36/512$ and $r_1 = 144/512$.

Strip Dominance: Given a polynomial $g(t) = a_0 + ... + a_n t^n$ we define

$$\langle g \rangle = \min(A_0, ..., A_n), \qquad A_k = a_0 + ... + a_k. \tag{18}$$

We have $g > 0$ on $[0,1]$ if $\langle g \rangle > 0$ and $g \geq 0$ on $[0,1]$ if $\langle g \rangle \geq 0$. Here is the proof:

$$g(t) = (1-t)(A_0 + A_1 t + \cdots + A_{n-1} t^{n-1}) + A_n t^n. \tag{19}$$

We call $G(t,h)$ *strip dominant* if there is a sign choice so that

$$\pm G(t,h) = g_0(t) + g_1(t)h + ... + g_m(t)h^m, \quad \langle g_0 \rangle > 0, \quad \min(\langle g_0 \rangle, ..., \langle g_m \rangle) \geq 0. \tag{20}$$

If G is strip dominant then G does not vanish on Ω. Here is a tweaked criterion: G does not vanish on Ω if $G(t/2,h)$ and $G(1-t/2,h)$ are strip dominant.

Certificate: To certify that f is nice on $\mathcal{M}|_{[x_0,x_1]}$ we do the following.

1. We choose r_0, r_1 as listed above, depending on the family.
2. We use the rational parametrization $F : \Omega \to \boldsymbol{R}^2$ such that $\mathcal{M}|_{[x_0,x_1]} \subset F(\Omega)$.
3. We write $f \circ F = P/Q$ where P and Q are integer polynomials.
4. We show that P and Q are either strip dominant or tweaked strip dominant.

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